

CHEM 165

LAB MANUAL

Spring 2000

Lab Schedule

Week 1	No Lab	
Week 2	Experiment 1:	Aspirin
Week 3	Experiment 2:	Iron Oxalate Complex
Week 4	Experiment 3:	Spectra of Copper Complexes
Week 5	Experiment 4:	Fremy's Salt
Week 6	Experiment 5:	Coordination Chemistry Part 1
Week 7	Experiment 5:	Coordination Chemistry Part 2
Week 8	Experiment 6:	Dissolved Oxygen

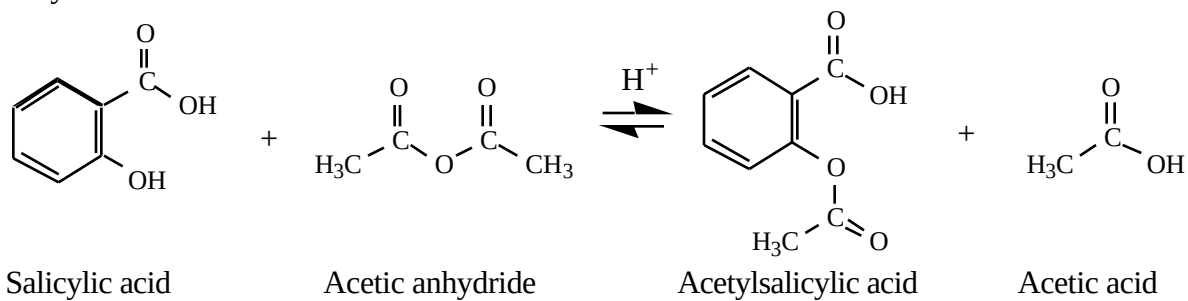
Reports are due one week after completion of the experiment.

EXPERIMENT 1

SYNTHESIS OF ASPIRIN

INTRODUCTION

Aspirin (acetyl salicylic acid) is to be prepared from a reaction between salicylic acid and acetic anhydride:



In the above reaction the hydroxy group (-OH) on the aromatic ring in salicylic acid reacts with acetic anhydride to form an ester functional group. This is also known as an esterification reaction.

There are two important aspects of this reaction that we must consider if we are to have a successful synthesis. The first is that esterification reactions must make use of an acid catalyst. Under neutral conditions salicylic acid and acetic anhydride are not very reactive. By themselves, they react to produce the corresponding ester, but only over a period of days. However, with an acid catalyst, the reaction rate is greatly accelerated and significant amounts of ester can be produced within a lab period.

If we started with equivalent amounts of the reactants, even with the acid catalyst the reaction would slow down towards the finish. We therefore use an excess of acetic anhydride, which speeds the consumption of salicylic acid. The excess anhydride is easily disposed of with water. Also, unreacted salicylic acid would be more difficult to deal with. See the “Procedure” section for more details.

The aspirin that you will prepare might not be very pure and should not be taken internally, even if the experiment gives you a headache. How pure is the aspirin you have synthesized? The most probable impurity is salicylic acid. You will determine % salicylic acid (%m/m) by visible spectrophotometry. Salicylic acid forms a highly colored complex with Fe(III) while aspirin does not form such a complex. By measuring the absorption (at **525 nm**) of a solution containing a known amount of aspirin in a solution containing excess Fe^{3+} , one can determine the % salicylic acid in the aspirin. A calibration graph will be provided for this purpose.

You will measure the melting point of a pure aspirin sample as well that of a 10% m/m salicylic acid. With aspirin as the solvent and salicylic acid as the solute in this 10% m/m sample, we can expect the melting point to be less than that of the pure aspirin. Given the % salicylic acid, you will calculate the molality of solute, $m_{\text{salicylic acid}}$. You will calculate ΔT from the difference in melting points between this sample and pure aspirin. Using these 2 values, you will estimate K_f , the molal freezing point depression constant, using the formula: $\Delta T = K_f m_{\text{solute}}$.

There are two reasons why we do not use our synthesized aspirin for the K_f determination. (1) Our aspirin is not sufficiently dry - there is still a considerable amount of water in the 'dried' sample and this would affect our results adversely. (2) The contamination levels are quite small (0.1 - 4%) meaning that we would be attempting to measure quite a small ΔT even if the sample was completely dry.

MATERIALS

From the Stockroom:

- 10-mL volumetric flask
- Micro-spatula
- 5-mL conical vial
- Spin vane
- Air condenser
- Hirsch funnel
- Plastic dish

From your Desk:

- Glass rod
- 2 beakers
- Aspirator trap bottle
- Small filter flask
- 2 cuvettes

Provided in the Lab:

- Chem-Anal spectrophotometers
- Acetic anhydride with 1-mL automatic dispenser bottles
- Acetylsalicylic acid (aspirin) in specimen bottles
- Phosphoric acid (85%) in dropper bottles
- 95% ethanol in 500 mL squeeze bottles
- 0.025 M $\text{Fe}(\text{NO}_3)_3$ in 0.5M HCl in 500 mL bottles
- Filter paper, small circles to fit Hirsch funnels
- Acetone in 500 mL squeeze bottles
- Melting point apparatus (3/section)
- Capillary tubes (for taking melting points)
- 150° red dye thermometer (from the TA's)
- Hot plate stirrer and aluminum block, 6/section (2 pairs have to share one)
- Disposable pipettes and bulbs
- Ringstands, clamps
- Heat Lamps
- Electronic balance

Salicylic acid in specimen

Waste Disposal:

- 4 liter glass waste bottle- LIQUID WASTE ONLY

Small glass bottle labeled "Waste Aspirin"- Solid aspirin waste
4 liter glass waste bottle- ACETONE WASTE ONLY
Trash cans- All paper including used filter paper circles

PROCEDURE

A. Synthesis of Aspirin

1. Clean the conical vial by rinsing with acetone (the rinse should be disposed of only in the 'Acetone Waste' bottle). It is important that the vial not have any water in it before starting the reaction because water will react with the acetic anhydride and decompose it.
2. Prepare a heating apparatus using an aluminum block, thermometer, and a hot plate. Adjust the temperature to 50°C . Weigh 0.21 g of salicylic acid (MW 138) and place this in a 5 mL conical vial. Then add 0.48 mL of acetic anhydride (Caution! Corrosive!) followed by two drops of 85% phosphoric acid (Caution! Corrosive!). Add a spin vane and attach an air condenser to the vial. Remove gloves and place them in the trash. Place this assembly in the aluminum block (see **Figure 1**). Stir the mixture. Once the solid dissolves stir the mixture an extra 3-4 minutes to ensure that the reaction has gone to completion.

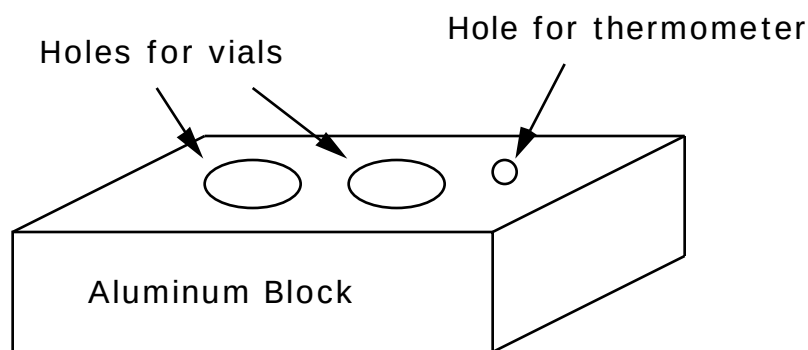


Figure 1

3. Remove the vial from the aluminum block and let it cool in an ice bath. While cooling, the aspirin may come out of solution. Once the solution has cooled, remove the air condenser and spin vane (with micro-spatula) and add 3 mL of water and stir. The water will decompose any excess acetic anhydride and will help precipitate any aspirin that remains in solution. The precipitate is isolated from the mixture by vacuum filtration. Set up a Hirsch funnel for vacuum filtration. Refer to **Figure 2** to make sure you have set it up correctly. The small filter flask must be firmly clamped or else the stiff rubber tubing will cause it to tip over. Weigh a small circle of dry filter paper, then place it in the Hirsch funnel. Moisten the filter paper with a few drops of water and turn on the aspirator. Transfer the mixture to the Hirsch funnel and filter off the water solvent. Add about 1 mL of cold water to the vial, stir, and transfer the mixture to the Hirsch funnel and filter. When all the precipitate has been collected in the funnel, rinse it with several 0.5 mL portions of water. Leave the product on the funnel for 5-10 minutes so it may air dry. Remove the filter paper bearing the precipitate and weigh it. Subtract the weight of the dry paper and record the mass of the crude aspirin. Calculate the percent yield.

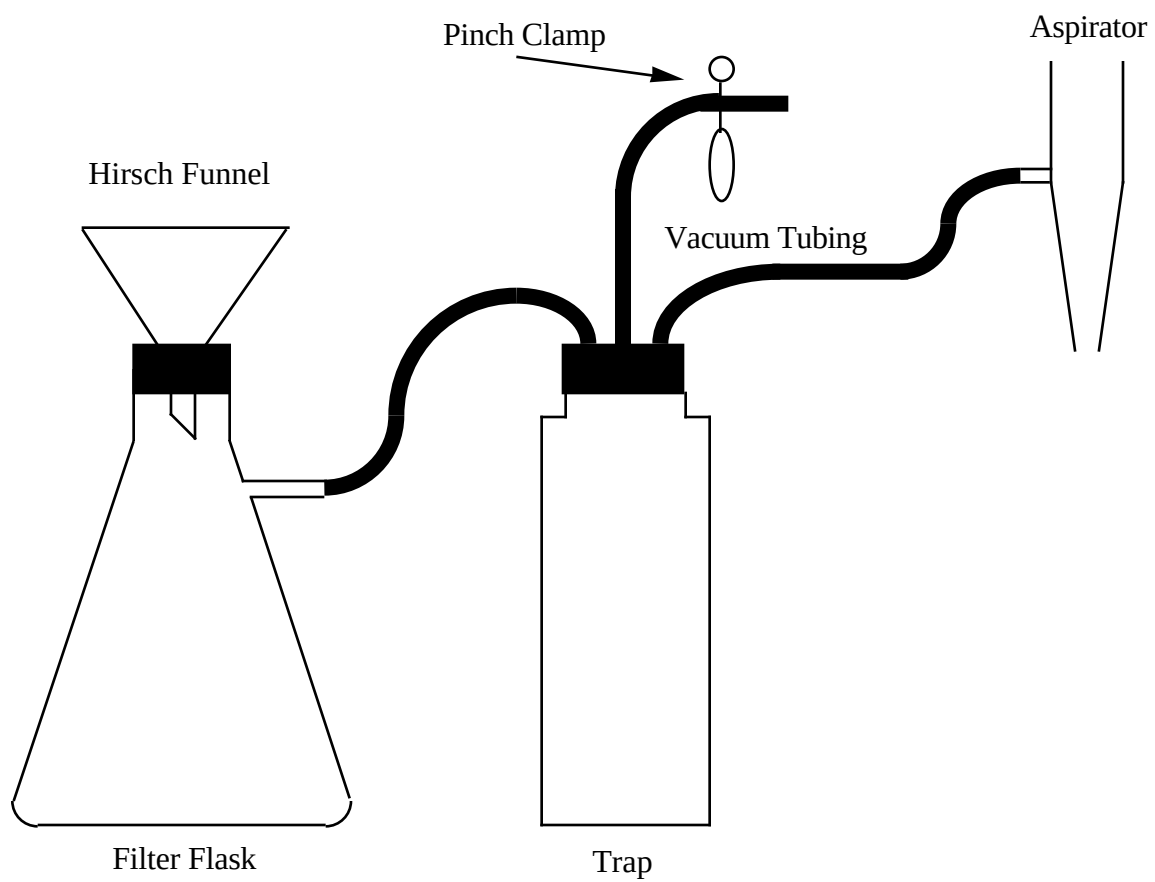


Figure 2

B. Determination of % Salicylic Acid

1. To analyze your crude aspirin for salicylic acid, weigh out between 0.020 and 0.023g of your sample on weighing paper. Record the exact mass. Carefully transfer this aspirin into a 10-mL volumetric flask. Dissolve the solid in 1 mL 95% ethanol. Add 1 mL of 0.025 M $\text{Fe}(\text{NO}_3)_3$ in 0.5M HCl and add distilled water to the 10 mL mark.
2. Rinse out one cuvette with a few 1-2 mL of the solution you just made (step #3) and fill the cuvette (3/4 full) with the same solution. Measure the absorbance at 525 nm on a Chem-Anal spectrophotometer. Refer to **Appendix D** for directions on the use of the spectrometers. **Figure 3** shows a typical spectrum for the Iron(III)-salicylic acid complex. The absorbance measurement should be made within 5 minutes of the time the sample was dissolved in ethanol, since aspirin will gradually decompose (i.e. hydrolyze) in solution, producing salicylic acid and acetic acid.
3. Calculate the % salicylic acid in the sample. The calibration graph (and equation) that you will need is given in **Figure 4**.

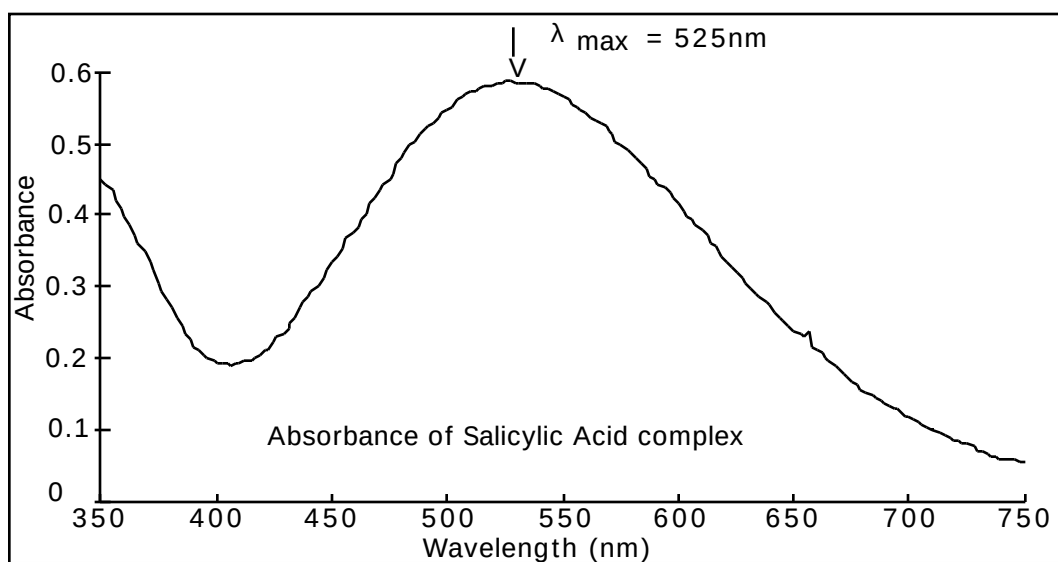


Figure 3

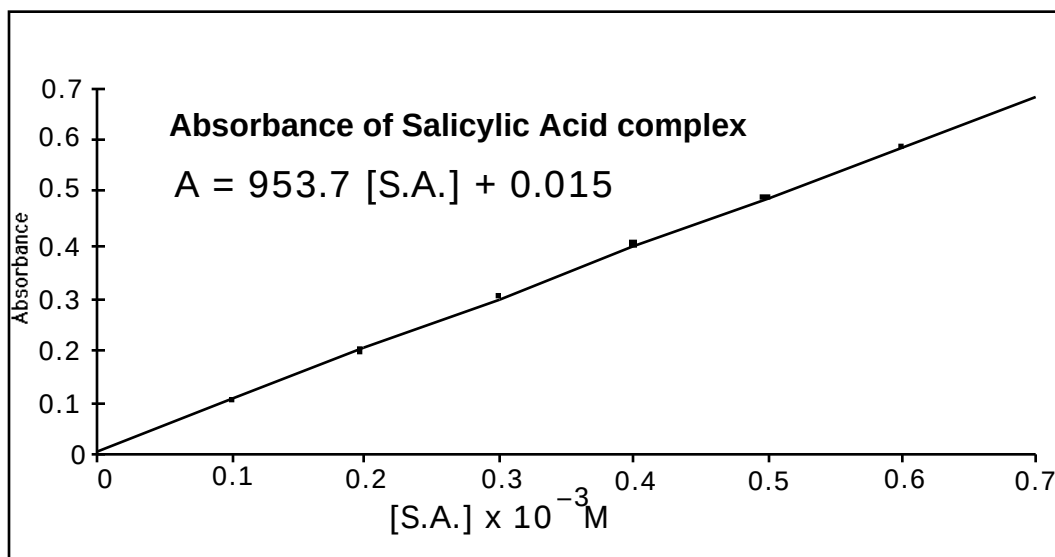


Figure 4

C. Melting Points and Determination of K_f

1. The melting point apparatus consists of a heated block with a voltage control to regulate the temperature rise, a light source, and a spy glass to watch the melting take place. A thermometer (-10 to 260°) is placed in its well and the samples are inserted in the block in tiny glass capillaries. Powder is pushed into the open end and the little tube (closed end down) is dropped down a long tube whose bottom rests on the counter. The shock packs the powder at the closed end of the capillary. Insert the sample tubes (both at once; your aspirin sample and the pure aspirin sample) in the block, heat slowly (~5°/minute), watch for melting, and read the thermometer. You will observe the powder melt over a range of temperature; record this range (the melting point) for each of the two samples. The block and light can become very hot, so keep your hands away.
2. From the 10 % salicylic acid in your sample, calculate $m_{\text{salicylic acid}}$ (molality of salicylic acid). From your measured melting points, calculate ΔT (change in melting point). From these two values, calculate K_f , the molal freezing point depression constant, using the formula : $\Delta T = K_f m_{\text{salicylic acid}}$. No literature value for K_f is available.
3. Put liquid waste into the 4 liter glass waste bottle. Solid aspirin goes into the waste aspirin jar. Before returning your conical vial to the stockroom, rinse it out with acetone with the rinse waste going into the 'Acetone Waste Only' bottle. All paper should be placed in the trash cans.

Questions:

1. Based on your crude aspirin mass, calculate the % yield. Comment on your % yield. What factors contribute to the yield being high or low?
2. If the density of acetic anhydride is 1.082 g/mL, prove it is indeed the reactant in excess.

EXPERIMENT 2

PREPARATION OF AN IRON OXALATE COMPLEX

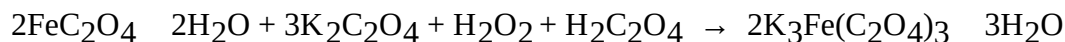
One of the more interesting aspects of inorganic chemistry is the study of large complex ions. The one studied here has iron as a central ion in the +3 oxidation state (also known as the ferric ion). Metal oxidation states are often written using Roman numerals, e.g. Fe(III) for the ferric ion. Such an ion can have six ligands, at the corners of an octahedron. An example would be six water molecules, or 6 OH⁻ ions etc. However, some ligands attach at two adjacent corners. The oxalate ion (⁻O₂C-CO₂⁻), minus at each end is such a bidentate ligand. Three of these ions bind to 3 pairs of adjacent corners.



The synthesis of potassium ferrioxalate (III), K₃Fe(C₂O₄)₃ · 3H₂O* (also called potassium ferrioxalate), involves two steps. The product of the first step is ferrous oxalate FeC₂O₄ · 2H₂O*, which is made by heating a solution of ferrous ammonium sulfate, Fe(NH₄)₂(SO₄)₂ · 6H₂O*, and oxalic acid, H₂C₂O₄. The balanced equation for the first step is



In the second step of the synthesis the ferrous oxalate from the first step is treated with hydrogen peroxide, H₂O₂, in the presence of oxalic acid and potassium oxalate, K₂C₂O₄, to give the complex ion [Fe(C₂O₄)₃]³⁻ (the oxalates are tightly bound to the Fe³⁺), which precipitates out of solution as the potassium salt, K₃Fe(C₂O₄)₃ · 3H₂O. The balanced equation for the second step is



* **Note:** The salts are hydrated. This means that there are water molecules bonded to the ions. The number of water molecules bonded to each salt 'unit' is indicated by the number after the dot: K₃Fe(C₂O₄)₃ · 3H₂O. These water molecules must be included in the molar mass of the salt.

The primary goal in a synthesis is to produce the desired product. A secondary consideration is to have as high a yield as possible. Yields are expressed as a percent of the theoretical yield, which is the total mass of product that is possible to make with quantities of reactants used.

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

The actual theoretical yields can be expressed in moles or grams. The theoretical yield is calculated from the amount of the limiting reagent, i.e. the reactant present in the smallest stoichiometric amount. Some syntheses have yields close to the maximum of 100%, while for others, a yield of 1% may be the best you can do.

In a synthesis involving more than one step, the overall yield depends on the yield of each step. If the first step has a yield of 80% and the second step has a yield of 90%, then the overall yield is 90% of 80%, which is 72%.

MATERIALS

From the stockroom:

Hot plates

From your desk:

3 small test tubes

Spatula

150 mL beaker

10 and 100 mL graduated cylinder

Thermometer

Glass stirring rod

Eye dropper

Filter flask

Buchner funnel

Aspirator trap bottle with cap and pinch clamp

Watch Glass

Provided in the lab:

Ferrous ammonium sulfate in specimen jars

1.0 M oxalic acid in 1 L bottles

Saturated potassium oxalate solution (30 g $K_2C_2O_4$ /100 mL soln=1.8M)
in 500 mL bottles

3% hydrogen peroxide (oxidizer) 3 g H_2O_2 /100 mL soln) in 1 pt bottles

95% ethanol (Flammable) in 500 mL squeeze bottles

7.5 cm filter paper

Waste Disposal: All waste is to be collected

4 L Amber bottle – liquid waste

32 oz. glass jar – the solid product

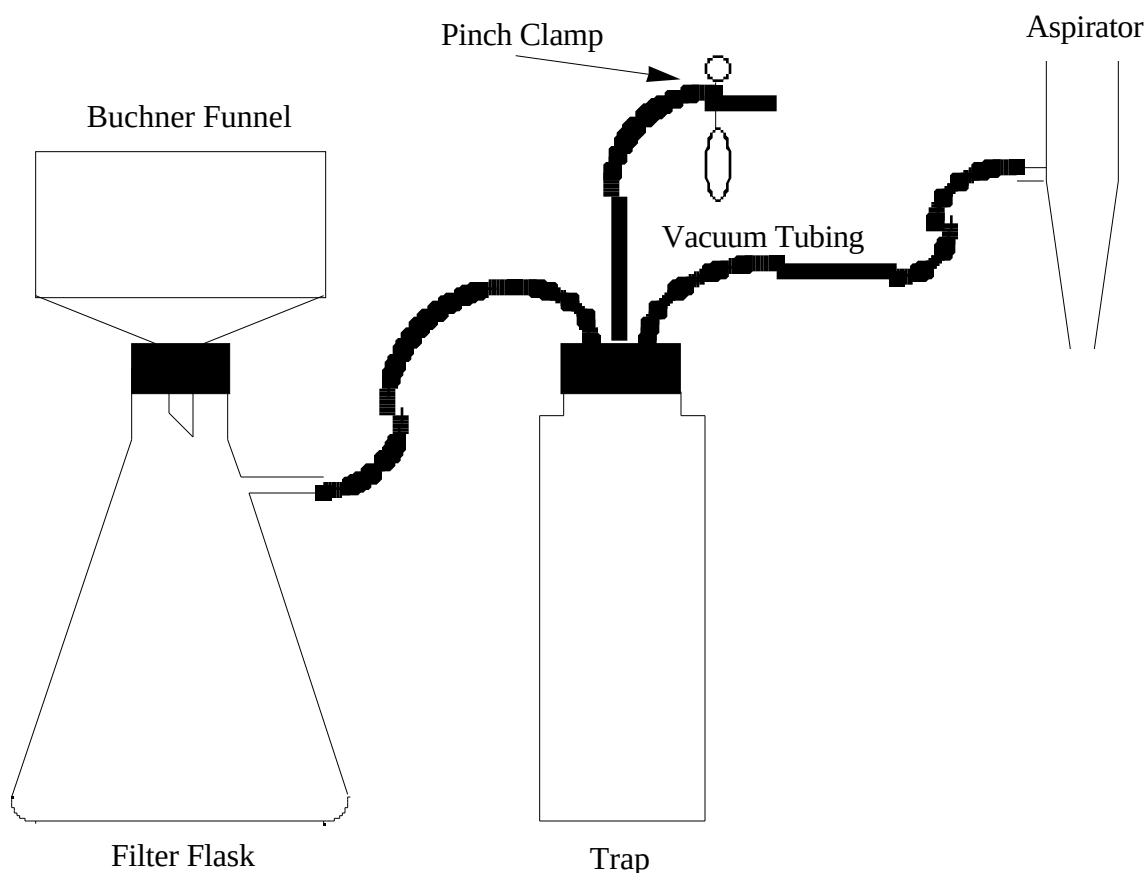
16 oz. glass jar – contaminated filter paper

PROCEDURE

Note: Your report on this experiment should include observations about color changes, precipitates, etc., and an explanation of each observation.

1. Weigh out about 3 g of ferrous ammonium sulfate on the electronic balance. Be sure to record the mass to the nearest milligram. Put the crystals in the 150 mL beaker and add 25 mL of deionized water. Stir well to dissolve.
2. Now add 25 mL of 1.0 M oxalic acid to the solution. Put the beaker on a hot plate and heat to boiling, stirring the mixture constantly.
3. Cool the mixture and allow the precipitate of ferrous oxalate to settle. Decant the liquid and wash the precipitate with 20 mL of deionized water. Warm the mixture to ca. 40°C and then allow the precipitate to settle again. Decant the wash liquid, removing as much as possible.
4. Add 10 mL of saturated potassium oxalate solution to the precipitate and warm the mixture to 40°C. Use a ring stand to hold the thermometer. **Slowly** add 20 mL of 3% hydrogen peroxide, stirring continuously while maintaining the temperature at 40°C (if the temperature rises above 50°C during this stage you must allow it to cool before adding the rest of the hydrogen peroxide). When all the hydrogen peroxide has been added, heat the mixture to boiling. Add 5 mL of 1.0 M oxalic acid all at once, then add 3 mL more dropwise. Keep the mixture near boiling. If the solution is not a clear green color, add up to 2 mL more oxalic acid until a clear green is obtained. If the solution is still slightly yellow-green, ask your TA about it. Continue boiling until the volume is reduced to 25-30 mL.
5. Turn off your hot plate and allow the solution to cool, then place in an ice bath. After 10 minutes, add 5 mL of cold ethanol (**Flammable!**) to aid the crystallization.
6. Allow the solution to stand undisturbed an additional 15 minutes while crystals are forming. While your product is crystallizing, set up a vacuum filtration apparatus as shown on the next page. The most common source of vacuum in the laboratory is the water aspirator. It consists of a water faucet that has a sidearm. The water flowing past the sidearm creates a reduced pressure due to the Bernoulli effect (just like the air flowing over an airplane wing). Secure a sidearm flask to a ringstand. To the sidearm attach one end of some thick, black vacuum tubing (thin amber tubing will collapse). Attach the other end to a trap. Then attach the trap (again using vacuum tubing) to the sidearm of a filter flask. In this way a partial vacuum will be on the filter flask. To the top of the filter flask attach a Buchner funnel. The flat bottom of the Buchner funnel is perforated. Write your name on a piece of filter paper in **pencil**, weigh it and then place it in the Buchner funnel. The filter paper keeps the solid (your product!) from being sucked in to the flask.

Note: The aspirator must always be used with a trap, as shown in the drawing on the next page. The reason is that the water pressure often fluctuates; if it drops, then the vacuum in the flask will suck in water from the aspirator. Without the trap, this water then goes into the filter flask, contaminating that solution (which you don't care about in this case, but might in other cases). The trap has a quick release valve so you can cut off the vacuum. If you see water starting to come into the trap, use the quick release valve to 'break' the vacuum and then turn off the water. Always use the quick release valve before you turn off the water!



7. Cool the solution in an ice bath for 10 - 20 minutes, then vacuum filter it. Simply turn on the water to the aspirator, dampen the (weighted) filter paper with a small amount of water, and then slowly transfer the contents of your beaker to the Buchner funnel. Once you've finished the transfer, keep the aspirator on for a few minutes. This will help dry the solid.
8. Wash and dry three test tubes. Label the first "water," the second "ethanol," and the third "50/50." Weigh out 0.3 g of your solid, dividing it into three equal pieces (by eye),

placing one piece in each test tube. Set aside for now, you will use them in a few minutes to do some solubility tests.

9. Weigh a watch glass and transfer your solid and filter paper to it. Calculate the mass of the solid by subtracting the mass of the filter paper and the watch glass from the total weight of the assembly. Set the watch glass under a heat lamp to dry your product.
10. While your product is drying, do the solubility tests. Add about 1 mL of deionized water to the first test tube, and swirl it to mix. After a couple of minutes check to see if the solid has completely dissolved. Record your result and set the test tube aside. **Repeat this procedure with 1 mL of ethanol and the** second test tube, and finally with 1 mL of a 50/50 solution (by volume) of water and ethanol added to the third test tube. Note that these results were obtained at room temperature. Set aside any test tube where the solid dissolved, then test the solubility of the other samples at a higher temperature. To do this, fill a beaker 1/3 full of water and heat it with a hot plate to 70–80°C. If sample did not completely dissolve at room temperature, heat the test tube in the hot water and swirl to mix. **Do not boil.** Record the temperature. Record whether the solid dissolves or not, noting that this test was done at a higher temperature. **When you have finished the solubility tests, pour the solutions into the waste container. If any solids remain, rinse them in to the waste container with water.**

Note: The next logical step after determining the solubility of your product in various solvents would be to use this information to purify it by ‘re-crystallization.’ Unfortunately there is not sufficient time in one lab period to do this. Your TA, however, will be glad to tell you how the procedure works.

11. Retrieve your watch glass from under the heat lamp and re-weigh the assembly, subtracting the mass of the filter paper and the watch glass to get the dry weight of your sample. Under normal circumstances you would use this mass to calculate the percent yield. But in this case you set aside 0.3 g of solid before it was dried. This mass must be included in the yield, but it is necessary to take into account the fact that the 0.3 g was not as dry as the heat-lamp dried product.
12. After the final weighing, dispose of your product in the 32 oz. glass waste jar. Put contaminated filter paper in the 16 oz. waste jar. All solutions should be disposed of in the 4 L amber waste bottle.

BEFORE LEAVING THE LABORATORY

1. Have your TA approve your calculations and check a sample of your product.
2. Clean your lab bench and have a TA check your equipment drawer, lab bench, and lab notebook.

Calculations

1. Residual solvent makes the yield appear larger than it actually is. The mass of the solid after drying under the heat lamp is less than the mass before drying; the ratio of the two masses can be used as a conversion factor from wet (residual solvent) to dry (no residual solvent) mass:

$$\frac{\text{mass after drying}}{\text{mass before drying}} \times 0.3 \text{ g} = \text{dry mass of product set aside}$$

The actual yield is obtained by adding the dry masses:

$$\text{mass after drying} + \text{dry mass of product set aside} = \text{total yield (in grams)}$$

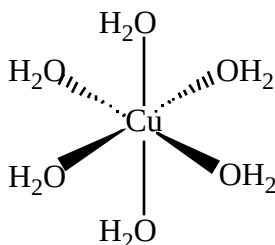
2. Show that iron is the limiting reactant throughout the synthesis.
3. Find the theoretical maximum yield in grams. Calculate the % yield from this result and your total dry yield in grams.

EXPERIMENT 3

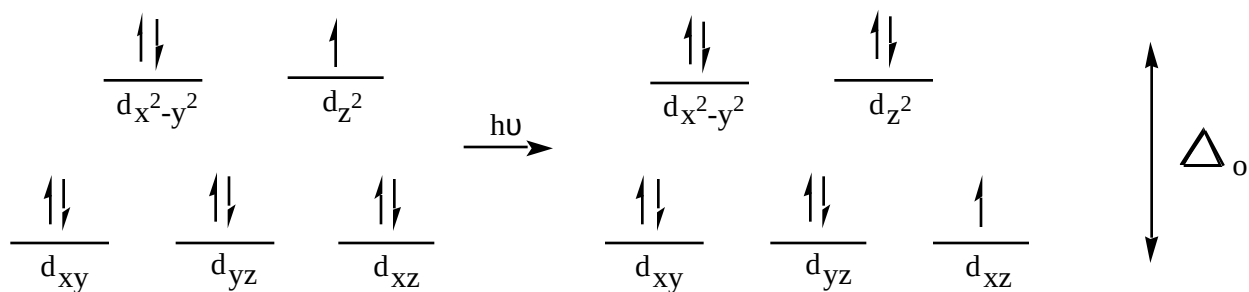
SPECTRA OF COPPER (II) COMPLEXES

INTRODUCTION

Copper (II) ions, when dissolved in water, exhibit a characteristic blue color. This observation is the result of two things. First, the copper ions are surrounded by six water molecules in an octahedron arrangement.



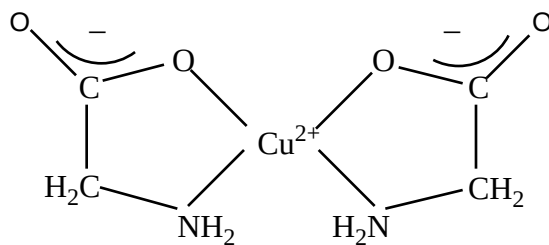
Second, the d orbitals of the transition element copper have one vacancy when it is in the +2 oxidation state. There is a crystal field splitting of the otherwise equal energy d orbitals that makes possible an electronic transition between the ground state and an excited state with the splitting Δ_o .



One purpose of this experiment is to correlate the Δ_o values with the spectrochemical series. It states that $\Delta_o(\text{Cl}^-) < \Delta_o(\text{H}_2\text{O}) < \Delta_o(\text{NH}_3)$ for the unidentate ligands in this experiment. There are no entries for the bidentate ligands.

In this experiment you will study three copper (II) ions with unidentate ligands and one with bidentate ligands. Ideally the bonding strength of the ligands should parallel the splitting Δ_o . However, none of the coordination complexes studied here are perfect regular octahedra. In the aquo complex (six waters) the two water molecules above and below the square plane are further from the central atom than the four in the square. In the ammine complex, the predominant species

has four NH_3 molecules in the square positions with waters above and below. The cis-Bis (glycinato) $\text{Cu}(\text{II})$ complex has the two glycine molecules in a particular square planar orientation as shown below.



$\text{Cu}(\text{gly})_2$

The CuCl_4^{2-} , unlike the others, has the four chlorides in a tetrahedral arrangement. These departures from the model may account for discrepancies in the spectra measured. In addition, the range of the automated recording spectrophotometer does not extend into the near infra-red, where much of the spectrum of the aquo and chloro complexes lie.

You will work in pairs to make the glycinate complex and separately in making the unidentate samples and spectra.

From your desk

Four 100 mL beakers
100 mL Graduated Cylinder
10 mL Graduated Cylinder
125 mL Filter Flask
Aspirator Trap
Spatula
Glass Stirring Rod
Grease Pencil
Glass Funnel
6 Test tubes

From the Stockroom

Crystallizing Dish
Small Buchner Funnel
Curvettes

Provided in the Lab

Hot Plate
Filter paper
Ice
Clay Pipe Triangle
Scanning spectrophotometers
Kimwipes
Copper sulfate pentahydrate
Glycine
Sodium bicarbonate
0.1 M Copper nitrate solution
Sodium Chloride
Concentrated Ammonium hydroxide
1 M HCl

Waste: All copper-containing solutions are to be collected in the amber waste bottles. Filter papers contaminated with copper complexes are to be collected in the plastic containers.

PROCEDURE

Be sure you record observations of color and appearance of solids and solutions during the reactions.

Preparation of cis-Bis (glycinato) copper (II) monohydrate, $\text{Cu}(\text{gly})_2 \cdot \text{H}_2\text{O}$

1. Place 1.0 g of copper (II) sulfate pentahydrate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, in a 100 mL beaker and add 6 mL of 1M HCl. The acid will keep the product complex in solution until you precipitate it in step 5.
2. When the complex has dissolved, add 0.5 g of glycine, $\text{NH}_2\text{CH}_2\text{CO}_2\text{H}$, to the solution.
3. Prepare a water bath by placing a clay pipe triangle in the crystallizing dish and filling the dish half full with tap water. This assembly is placed on a hot plate and the beaker containing the solution is placed in the water so that it rests on the clay pipe triangle.
4. Warm the solution for one hour with the hot plate on low. During this hour, proceed to steps 8 through 11 (preparation and spectra of the unidentate complexes).
5. Add sodium hydrogen carbonate, NaHCO_3 , in small portions (avoid a large excess) until precipitation is complete and CO_2 evolution stops. Do not add water or rinse the filtrate with water, as it will substantially decrease your yield. Suction filter the precipitate with a Buchner funnel.
6. Place the solid in a 100 mL beaker. Add about 40 mL of deionized water and place in the hot water bath, swirling the beaker frequently. Set the hot plate on 4.
7. When the $\text{Cu}(\text{gly})_2 \cdot \text{H}_2\text{O}$ solid has dissolved, some solid impurities may remain. Support the conical glass funnel with an iron ring and position it over a 100 mL beaker. Select a filter paper of appropriate size; fold it in quarters and open the fold to form a paper cone filter. Place it in the funnel (it should come up about 2/3 of the glass wall) and moisten with a little deionized water to seat it properly. Then carefully pour the hot solution through this “gravity” filter. Preserve the liquid filtrate. This solution will be used in the spectrum measurement.

Preparation of the Unidentate Samples

8. Obtain 20 mL of the stock copper nitrate solution. Observe and record the color of this aquo (water) complex. Reserve 10 mL for spectrum measurement.

9. To the other 10 mL of copper nitrate solution add 10 mL of deionized water and 4 g of solid NaCl and stir until dissolved. Record the color of this chloro (Cl^-) complex. Dilute a few mL of this complex with water in a test tube. Record any color change (is it still chloro complex?). Take 10 mL of the chloro complex to use in the ammine reaction and reserve the rest for spectrum measurement.
10. To the 10 mL of chloro complex solution add 1 mL concentrated NH_4OH (do this in the hood!). Observe the result. Initially, light blue copper hydroxide precipitates. This should re-dissolve upon stirring. If necessary, add more NH_4OH . Reserve for spectrum measurements.
11. Take spectra of your four complexes (see Appendix for operation of the HP diode-array spectrophotometer). With planning, you can do the three unidentate ones while the bidentate complex is “cooking.” Copies of spectra for $[\text{Cu}(\text{en})_2(\text{H}_2\text{O})_2]^{2+}$ and $\text{Cu}(\text{acac})_2$, where en is ethylenediamine and acac is the acetylacetonate ion, are attached.
12. Copper solutions are collected in bottles, Filter paper is collected in jars.

Calculations and Results

1. Calculate the moles of copper and glycine and HCl in your synthesis reaction. Which reagent, copper or glycine, is present in excess and by what percent?
2. If you assume 100% yield, what would be the approximate concentration of the complex in moles per liter in the filtrate obtained in step 7?
3. A spectrochemical series is an ordering of similar complexes on the basis of splitting energy. Remembering that

$$E = h\nu = \frac{hc}{\lambda}$$

for one photon of light energy (where λ is the wavelength corresponding to maximum absorbance), calculate the splitting energy of the six complexes for which you have spectra and determine the spectrochemical series for these complexes. Be sure to indicate whether your ranking is low to high, or high to low. Note: The distance units of c and λ must match in order to cancel them out. Give E in joules. It will be per molecule and thus seem small. Multiply by Avogadro's number to get joules per mole of complex.

4. For each complex, indicate what color is absorbed and what color is observed. Compare with your observations.

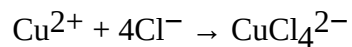
APPENDIX D: USE OF THE DIODE ARRAY SPECTROPHOTOMETER

A SoftKey is a key marked with an F and a number. F0 on the program corresponds to the F10 key.

1. You want to start at General Scanning. If the program is at another level, you can change it to General Scanning by using the F10 key.
2. The beam from the spectrophotometer goes from the left to right as you face the computer screen, so orient your cuvette so the 'best sides' are at 90° rather than facing you. Lift the lever on the cuvette holder and place the properly aligned reference cuvette in the slot and push the lever gently back into place.
3. Press F8 to scan the reference (also called a blank). You will see a spectrum of your reference (deionized water) on the screen. The computer will store this spectrum and automatically subtract it from your spectra.
4. Lift the lever on the cuvette holder, remove the reference, and place a sample in the holder. Lower the lever and press F1 key to scan. The sample spectrum (with reference already subtracted) is displayed.
5. Press F8 (Sample Info), to enter the sample name: aquo, chloro, amino, or glycinato complex. Press enter until you get a 'press a SoftKey' prompt.
6. Press F2 and use the keyboard arrows to position the screen arrow at the peak maximum, then press F1 and the wavelength and the absorbance of that point will be included with your spectrum.
7. Press F10 to exit cursor mode and then press F9 to get a printout of your spectrum and data. To run another sample, repeat from step 4 with the new sample. If you are finished, lift the lever on the cuvette holder and remove your cuvette. Return the program to General Scanning by pressing F10 until you are back to General Scanning.

QUESTIONS

- 1.) Correlate your values of Δ_o with the spectrochemical series.
- 2.) Interpolate the Δ_o values for bidentate ligands in your unidentate series (Question 1.).
- 3.) Write out equilibrium constants for K (complex) based on the equations for formation.
Example:



$[\text{Cu}^{2+}]$ is concentration of aqua complex.

EXPERIMENT 4

PREPARATION OF A STABLE FREE RADICAL FREMY'S SALT AND ITS TETRAPHENYLARSONIUM DERIVATIVE

Note: This experiment must be completed without a break during one period, and the instructions must be precisely followed, or the result is frustration and/or despair.

The compound prepared in this experiment is an odd molecule, that is, it contains an odd number of electrons. The salt potassium nitrosulphonate $[K_2^+(ON(SO_3)_2)^{-2}]$ was first reported by Fremy in 1845 and has the common name Fremy's salt. The solid has an orange-yellow color but it gives purple solutions. Recent work has shown that some other salts of the $ON(SO_3)_2^{-2}$ anion give purple solids.

Dilute solutions of Fremy's salt have been studied by electron spin resonance spectroscopy and are sometimes used as standards in this field of spectroscopy.

MATERIALS

From the Stockroom:

- 250 mL Erlenmeyer flask
- Spatula
- (2) 150 mL beaker
- 250 mL beaker
- 800 mL beaker (for ice bath)
- 100 mL graduated cylinder
- Thermometer
- Glass stirring rod
- Filter flask
- Buchner funnel
- Aspirator trap bottle with cap and pinch clamp
- Rubber policeman
- (3) Long-stemmed funnels

Provided in the Lab:

- Potassium nitrite
- Potassium acetate
- Potassium permanganate
- Potassium chloride

Sulfur dioxide (lecture bottles)
Concentrated ammonium hydroxide
NaCl (for ice bath)
Ether (Caution: Flammable)
Tetraphenylarsonium chloride
95% ethanol (Caution: Flammable) in 500 mL squeeze bottles
Fluted filter paper
7.5 cm filter paper
Water trough
Sample vials
Funnel holder
pH paper

Waste Disposal: All waste is to be collected.

Part A: 4 L Amber bottle- liquid waste (filtrate)

Part B: 4 L Amber bottle- liquid waste (filtrate)
16 oz glass jar- contaminated filter paper with MnO_2
16 oz glass jar- Fremy's salt product

Part C: 4 L Amber bottle- liquid waste (filtrate)
16 oz glass jar- contaminated filter paper with

a) Preparation of potassium hydroxylamine disulphonate

Dissolve 6 g potassium nitrate and 9 g potassium acetate in 60 mL of water in a 250 mL flask and cool in an ice-salt freezing mixture to about -5°C . Add 100 g of finely crushed ice, shaking frequently. Bubble sulphur dioxide slowly through the solution. A mass of crystals of potassium hydroxylamine disulphonate, $\text{HO}-\text{N}(\text{SO}_3\text{K})_2 \cdot 2\text{H}_2\text{O}$ crystallizes out. When the odor of SO_2 shows that an excess has been passed (*ca.* 15 min.), filter off these crystals (**fume hood**) under suction and wash them with 30 mL ice cold water. Keep the crystals for the next step.

b) Preparation of Fremy's salt

Prepare a solution of 2 g KMnO_4 in 70 mL of water and cool in ice. Dissolve the potassium hydroxylamine disulphonate prepared in the previous section in a minimum amount of (about 100 mL) water at room temperature. Add more water 10 mL at a time with stirring until clear. Wait 2 minutes between additions. Make this solution basic (pH 8-9, check with pH paper) through the addition of a few drops of concentrated ammonia, then add the cold potassium permanganate. Stir vigorously and quickly remove the thick precipitate of MnO_2 which forms, by gravity filtering the suspension through three large funnels fitted with fluted filter papers.* The filtrate is allowed to come to room temperature, however the unfiltered suspension is kept in an ice bath. A total of approximately 35 mL of pink filtrate is collected. To the combined filtrate add, with stirring, small amounts of solid KCl until Fremy's salt starts to separate as a crystalline precipitate of yellow needles. Cool in ice and wait a few minutes until crystallization is complete. Filter at the water pump and wash with a little ice water. Then wash with similar volumes of ethanol and finally acetone and air dry the product.

The compound is yellow and diamagnetic in the solid state but dissolves in water to give a violet, paramagnetic solution. The solution contains the free radical $\text{ON}(\text{SO}_3)_2^{-2}$ which dimerizes in the solid. How would you explain the bonding in this compound?

The product will keep almost indefinitely if very pure, but most preparations decompose suddenly (with a feeble and harmless explosion) after a few hours. Show product to instructor.

c) Preparation of Tetraphenylarsonium Salt

Add solid potassium nitrosyldisulfonate from b) (use about 0.1 g of your product) to an aqueous solution of Tetraphenylarsonium chloride (0.1 g) in water (5 mL). Stir the solution vigorously and cool in ice. Wash with ice water and ether and air dry the solid. Hand in this product to the instructor in a small sample vial.

* Three filters are used to speed the process.

QUESTIONS

1. Write balanced chemical equations for the reactions occurring in parts (a) and (b).
2. Explain the bonding in the $\text{ON}(\text{SO}_3)_2^{-2}$ ion.
3. Explain the difference in color between the solid and solutions of $\text{K}_2\text{ON}(\text{SO}_3)_2$. Explain the difference in color between the solids $[(\text{C}_6\text{H}_5)_4\text{As}]_2\text{ON}(\text{SO}_3)_2$ and $\text{K}_2\text{ON}(\text{SO}_3)_2$.

References

1. Cotton, FA and Wilkinson, G. *Advanced Inorganic Chemistry*, 3rd Ed. Interscience; 1972: p. 340.
2. Filmore, RL and Wilson BJ. *Inorganic Chemistry* 7. 1968; 1592.
3. Moser, W et al. *J Chem Soc.* 1968; 3039:3043.
4. Cottrell, WRT and Farrar, J. *J Chem Soc. (A)* 1970;1418.
5. Moser & Howie, *J. Chem. Soc. (A)* 1968, 3039-3047.
6. Cottrell & Farrar, *J. Chem. Soc. (A)* 1979, 1418-1420.

EXPERIMENT 5

COORDINATION CHEMISTRY: Isomerization of *cis*-Dichloro bis (1,2 - diaminoethane) cobalt(III) chloride a.k.a. (*cis*-Dichlorobis (ethylenediamine) cobalt(III) Chloride)

INTRODUCTION

Complexes of cobalt(III) are numerous (as are those of Cr(III)) largely because although these may often be thermodynamically unstable in aqueous solution with respect to corresponding aqua-complexes, the rates of aquation, or of replacement of one ligand by another, are usually very low. Thus different complexes may result from different syntheses. Care must be taken to follow such recipes closely, otherwise unwanted complexes may result, which are equally reluctant to convert to the desired product.

Again, because of their kinetic stability, it is possible often to obtain isomeric forms where these exist. In this experiment, you will synthesize the isomeric *trans*- and *cis*- forms of $[\text{Co}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_2\text{Cl}_2]\text{Cl}$. You will then determine the rate of constant k for the conversion of the *cis*-isomer to *trans*-isomer by following the disappearance of the visible (electronic) absorption of the *cis*-isomer.

MATERIALS

From the Stockroom:

- 400 mL beaker
- Spatula
- mortar and pestle
- 10 mL graduated cylinder
- 100 mL graduated cylinder
- Thermometer
- Glass stirring rod
- Filter flask
- Buchner funnel
- Aspirator trap bottle with cap and pinch clamp
- Watch Glass
- Evaporating dish
- 100 mL volumetric flask with cap
- 250 mL volumetric flask with cap

1 cm cuvette (cell)
Hot plate/magnetic stirrer
Stir bar

Provided in the Lab:

Ethylenediamine
Cobalt chloride hexahydrate
30% hydrogen peroxide
Concentrated hydrochloric acid
Ether (Caution: Flammable)
95% Methanol (Caution: Flammable) in 500 mL squeeze bottles
Acetone
7.5 cm filter paper
Water trough
Thermostatic water bath (set at 40 degrees C)
HP Diode-array spectrophotometer

Waste Disposal: All waste is to be collected.

4 L Amber bottle- liquid waste (filtrate)
16 oz glass jar- contaminated filter paper
16 oz glass jar- Cobalt complex products

PROCEDURE

Safety Reminder: 1,2-diaminoethane (ethylenediamine) will burn skin, vapor harmful.

Preparation of Trans-[Co(en)₂Cl₂]Cl (en = 2,3-diaminoethane)

Grind CoCl₂ · 6H₂O (8 g) finely and dissolve in water (25 mL) in a 400 mL beaker. Work in the hood from here on. Prepare 30 mL of 10% aqueous 1,2-diaminoethane (ethylenediamine) solution and add it to the cobalt solution, while stirring with a magnetic stirrer.

Cool in ice water and keeping cold, add, 1–2 mL at a time, 15 mL of 30% hydrogen peroxide, all the time stirring magnetically. Hydrogen peroxide is a powerful oxidizing agent and must be used with extreme care. When the addition is complete and effervescence (what is the gas?) has subsided, carefully add concentrated HCl (17 mL) while stirring. Using a large beaker, heating on the hotplate (set on high), concentrate the solution until green crystals form over the surface. It is important that the solution does not become too concentrated.

Allow the now concentrated solution to cool in an ice bath for 30 min. (Set up steam bath while waiting). Filter off (vacuum filter) the green square crystals of trans-[Co(en)₂Cl₂]Cl · HCl. Wash with cold methanol and ether. Seal a small portion in an ampoule. Dry the remainder in the oven at 110°C for at least 2 hours then grind two or three times in cold methanol to remove the HCl. Wash with ether and suck dry. Weigh and store in the desiccator.

Cis-[Co(en)₂Cl₂]Cl

Dissolve half of the dried trans-isomer in the minimum amount of hot water (at 90°C; this requires about 4 mL of water of every gram of the trans-isomer) in a small evaporating dish. Evaporate the solution to near dryness on the steam bath (use a beaker of water on bench) with constant stirring (glass rod). Repeat the procedure. The purple product obtained is the cis-isomer. Allow to cool, scrape into a Hirsch or Buchner funnel. Wash with acetone and suck dry. Give yield and % yield for each isomer.

Measurement of Visible (Electronic) Absorption Spectra and Rate Constant for Isomerization

The operation of UV-visible spectrometer will be demonstrated.

Prepare accurately 2.5×10^{-3} M solutions (50 mL) of each isomer by first grinding the solid in methanol and then diluting. Record the UV-visible spectrum of the two solutions over the range 350–700 nm in 1 cm cells. Be sure to first run a baseline for each spectrum, with methanol in both cells. Determine the extinction coefficients at the peaks. Notice that the band maximum at 540 nm for the cis coincides with a minimum in the spectrum of the trans-isomer and thus is a convenient point to follow the cis to trans isomerization. Print out the spectra of your pure cis and pure trans samples. Use the pure trans sample for A_{∞} .

Immerse the solution of the cis-isomer in the thermostatically controlled water bath at 40°C. After equilibration, withdraw samples every ten minutes, cool rapidly in an ice bath and determine the absorbance at 540 nm using the Coleman (see instructor). Leave a sample in the bath overnight and then determine its absorbance to obtain A_{∞} . Plot $\log (A_t - A_{\infty})$ vs. t and determine the rate constant k for the reaction in units of min^{-1} .

Theory: During the isomerization of cis to trans the UV-visible absorbance (A_t) of the solution at a given wavelength at time t is given by the Beer-Lambert law as:

$$A_t = \epsilon_1 c_1 l + \epsilon_2 c_2 l$$

where c_1 is the concentration and ϵ_1 the molar extinction coefficient at that wavelength of the cis-form; c_2 and ϵ_2 are the analogous quantities for the trans-isomer. l is the pathlength of the solution.

At the completion of the reaction ($t = \infty$) the cis-form has been completely converted to the trans-isomer; if we call the concentration of the trans-isomer at this stage c_0 then

$$A_{\infty} = \epsilon_1 c_0 l$$

But from the stoichiometry of the reaction

$$c_0 = c_1 + c_2$$

$$\text{i.e. } A_{\infty} = \epsilon_2 (c_1 + c_2) l$$

from which $\epsilon_2 c_2 l = A_{\infty} - \epsilon_2 c_1 l$. Substituting this relationship in to equation (1),

$$A_t = \epsilon_1 c_1 l + A_\infty - \epsilon_2 c_1 l$$

$$\text{from which } c_1 = \frac{A_t - A_\infty}{l(\epsilon_1 - \epsilon_2)} = a(A_t - A_\infty)$$

where a is a constant.

The first order rate equation for the disappearance of the cis-isomer:

$$\log c_1 = \frac{-k}{2.303} t + \text{constant}$$

(k is a rate constant)

$$\log(A_t - A_\infty) = \frac{-k}{2.303} t + \text{constant}$$

(log a is absorbed into the const.)

Thus a plot of $\log(A_t - A_\infty)$ vs. time should be a straight line with slope

$$\frac{-k}{2.303}$$

DISCUSSION

Write balanced equations for the synthesis of the trans-isomer and calculate theoretical and % yields for both isomers. Explain the purpose of the hydrogen peroxide in the preparation of trans-[Co(en)₂Cl₂]Cl and give appropriate equations. Sketch the structures for the two isomers, including enantiomorphs.

Compare your spectra with the literature (see Cotton and Wilkinson) and assign the absorption bands using the theoretical energy level diagrams for the splitting of the d-orbital energy levels. Comment on the values of the extinction coefficients.

Simple salts of cobalt are almost all cobalt (II) (e.g., $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$), whereas complexes with N-donor ligands (like en) are almost all low spin cobalt (III). Suggest why this is so.

References

Leverett, P and Oliver, MJ. J Chem Ed. 1976; 53: 440

EXPERIMENT 6

ANALYSIS OF WATER FOR DISSOLVED OXYGEN

To gain a basic understanding of quantitative techniques of volumetric analysis by determining the dissolved-oxygen content of a water sample.

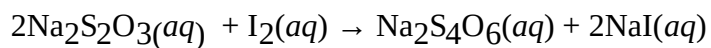
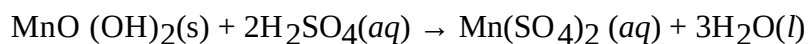
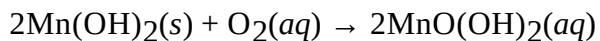
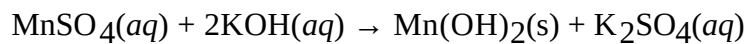
INTRODUCTION

The oxygen normally dissolved in water is indispensable to fish and other water-dwelling organisms. Certain pollutants deplete the dissolved oxygen during the course of their decomposition. This is particularly true of many organic compounds that are present in sewage or dead algae. These are decomposed by the aerobic metabolism of microorganisms, which use these organic compounds for food. The metabolic process is an oxidation of the organic compounds $\frac{3}{4}$ the dissolved oxygen is the oxidizing agent. Thus while these microorganisms are removing the pollutants, they are also removing the dissolved oxygen that otherwise would be present to support aquatic life. Since the solubility of most gases in solution decreases as the temperature of the solution increases, thermal pollution also decreases the dissolved oxygen content.

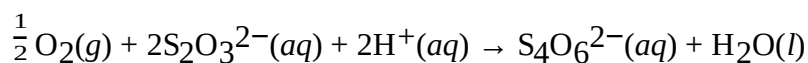
As a logical consequence of this, one empirical standard for determining water quality is the dissolved-oxygen content (DO). The survival of aquatic life depends upon the water's ability to maintain certain minimum concentrations of the vital dissolved oxygen. Fish require the highest levels, invertebrates lower levels, and bacteria the least. For a diversified warm-water biota, including game fish, the DO concentration should be at least 5 mg/L (5 ppm). Another water quality standard is the biological oxygen demand (BOD). The BOD is the amount of oxygen needed by the microorganism to remove the pollutant. In order to determine BOD, a sample containing organic pollutants is incubated with its microorganisms for a definite time, usually 5 days, and the amount of oxygen removed is measured. The BOD is taken as the difference in DO before and after incubation. A BOD of 1 ppm is characteristic of nearly pure water. Water is considered fairly pure with a BOD of 3 ppm, and of doubtful purity when the BOD level reaches 5 ppm. Monitoring of water quality therefore logically includes analysis of dissolved oxygen.

This experiment outlines the analysis of water samples for their dissolved oxygen (DO) content using the azide modification of the iodometric (Winkler) method. This is the procedure most commonly used for analysis of sewage, effluents, and streams. It is based on the use of manganous compounds that are oxidized to manganic compounds by the oxygen in the water sample. The manganic compound in turn reacts with KI to produce iodine, I_2 . The released I_2 is

then titrated with standardized sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, using starch as an indicator. The chemical reactions involved are as follows:



The net overall chemical equation for this sequence of reactions is:



MATERIALS

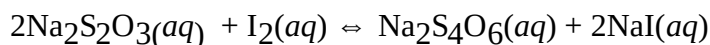
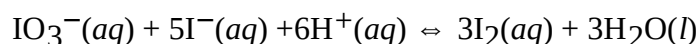
Balance
Thermometer
50 mL buret
250 mL Erlenmeyer flasks (3)
500 mL Erlenmeyer flask
250 mL narrow-mouth, glass-stoppered bottle or 1 pt bottle
2 mL pipets (2)
25 mL pipet
1 L volumetric flask
250 mL volumetric flask
KI, NaN_3 , MnSO_4 , NaOH, KIO_3
conc. H_2SO_4
water sample (unknown)
chloroform
1% boiled starch solution
 $\text{Na}_2\text{S}_2\text{O}_3$ solution (2 g $\text{Na}_2\text{S}_2\text{O}_3$ /100 mL H_2O)
alkaline iodine-azide solution
2.15 M MnSO_4 (freshly prepared)
2 NH_2SO_4

PROCEDURE

a) Standard Sodium Thiosulfate Solution

First prepare and standardize a 0.025 N sodium thiosulfate solution as follows: Begin to boil about 600 mL of distilled water. Once it boils, remove from heat. Weigh out about 3 g of $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ and place in 500 mL volumetric flask and dissolve it. Add 0.2 g of NaOH to retard bacterial decomposition and dilute to 500 mL in a 1 L volumetric flask, using previously boiled water.

Accurately weigh out about 0.1 g of potassium iodate in a 250 mL volumetric flask and place in a 100 mL volumetric. Make up to the mark with DI water. To each of three 25 mL aliquots of this solution in labeled 250 mL Erlenmeyer flasks add 0.5 g of potassium iodide and about 2 mL of 2 N sulfuric acid. Titrate each of these solutions with your thiosulfate solution, with constant stirring. When the color of the solution has become a pale yellow, dilute to approximately 200 mL with distilled water, add about 2 mL of 1% starch solution, and continue the titration until the color changes from blue to colorless for the first time. Ignore any return of color. Record the final buret reading and subtract that value from the initial reading to give the amount of thiosulfate used. Potassium iodate has an equivalent weight of 35.67 g/equiv. The reactions involved in this standardization are



You are actually titrating with your thiosulfate the iodine formed by the first reaction. Calculate the normality of your thiosulfate from the following equations:

$$\text{Equivalents Na}_2\text{S}_2\text{O}_3 = \text{equivalents KIO}_3$$

$$\text{Equivalents Na}_2\text{S}_2\text{O}_3 = V_{\text{Na}_2\text{S}_2\text{O}_3} \times N_{\text{Na}_2\text{S}_2\text{O}_3}$$

$$\text{Equivalents KIO}_3 = \frac{(\text{g KIO}_3)}{(35.67 \text{ g/equiv})} \times \frac{25.00 \text{ mL}}{250.0 \text{ mL}}$$

Thus,

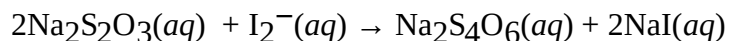
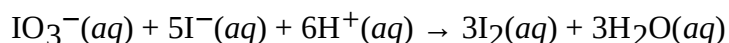
$$N_{\text{Na}_2\text{S}_2\text{O}_3} = \frac{(\text{g KIO}_3)(25 \text{ mL})}{(35.67 \text{ g/equiv}) (\text{volume Na}_2\text{S}_2\text{O}_3 \text{ in liters}) (250 \text{ mL})}$$

Example 1

A 0.2264 g sample of KIO_3 was dissolved in 250.0 mL of water. To each 25.00 mL aliquot of this solution were added 0.500 g of KI and 2.00 mL of 2.00 NH_2SO_4 . Titration of the liberated iodine required 25.30 mL of $\text{Na}_2\text{S}_2\text{O}_3$ solution. Calculate the normality of the $\text{Na}_2\text{S}_2\text{O}_3$ solution.

Solution:

From the analytical reactions:



it is seen that exactly 6 equiv of iodine is liberated for each mole of iodate. Thus KIO_3 (MW = 214.006) has as equivalent weight of

$$\frac{214.006 \text{ g/mol}}{6 \text{ equiv/mol}} = 35.67 \text{ g/equiv}$$

whence

$$\begin{aligned} \text{Equivalents } \text{KIO}_3 &= \frac{0.02264 \text{ g}}{35.67 \text{ g/equiv}} \\ &= 6.347 \times 10^{-4} \text{ equiv} \end{aligned}$$

and since

$$\begin{aligned} \text{Equivalents } \text{Na}_2\text{S}_2\text{O}_3 &= \text{Equivalents } \text{KIO}_3 \\ &= 6.347 \times 10^{-4} \text{ equiv} \end{aligned}$$

then

$$N_{\text{Na}_2\text{S}_2\text{O}_3} = \frac{\text{equiv } \text{Na}_2\text{S}_2\text{O}_3}{\text{liters of solution}}$$

$$= \frac{6.347 \times 10^{-4} \text{ equiv}}{0.02530 \text{ L}}$$

$$= 0.02509 \text{ N}$$

b) Water Sample Analysis

Collection of Sample

Collect the sample in a narrow-mouthed, glass-stoppered bottle (250–300 mL capacity). Avoid entrapment or dissolution of atmospheric oxygen. Allow the bottle to overflow its volume and replace the stopper so that no air bubbles are entrained; avoid excess agitation, which will dissolve atmospheric oxygen. Record the temperature of the water sample in degrees Celsius. The sample should be analyzed as soon as possible.

Release of Iodine

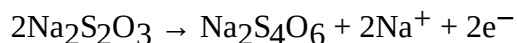
Open the sample bottle with great care to avoid aeration and add 0.25 g NaOH. Allow to dissolve. Add 0.5 g MnSO_4 and 0.25 g KI. Thoroughly mix the contents of the bottle by inverting the bottle several times. A milky precipitate forms and gradually changes to a yellowish-brown color. Allow the precipitate to settle so that the clear solution occupies the top third of the bottle. Carefully remove the stopper and immediately add 2 mL of concentrated sulfuric acid. This addition should be made by bringing the pipet tip against the neck of the bottle just slightly below the surface of the liquid. Stopper the bottle and then mix the contents by gentle inversion until the precipitate dissolves. At this point the yellowish-brown color due to liberated iodine should appear. The sample need not be titrated immediately, but if titration is delayed, the sample should be stored in darkness. The titration should be done within several hours.

Titration

Measure accurately 50 mL of the sample into a beaker. Titrate the sample with the standardized thiosulfate solution with constant stirring. When the color of the solution becomes a pale yellow, add about 2 mL of 1% starch solution and continue titrating until the color changes from blue to colorless for the first time. Record the volume of titrant necessary. Analyze your second sample.

Calculation of Dissolved Oxygen Content

Because $\text{Na}_2\text{S}_2\text{O}_3$ undergoes a one-electron change in its reaction with iodine, a 0.025 N solution is also 0.025 M:



According to the above equations, 1 mol of O_2 (32 g) requires 4 mol of $\text{Na}_2\text{S}_2\text{O}_3$ in reaching an end point. The number of moles of $\text{Na}_2\text{S}_2\text{O}_3$ is equal to the volume of $\text{Na}_2\text{S}_2\text{O}_3$ (in liters) times the concentration of the $\text{Na}_2\text{S}_2\text{O}_3$ (in molarity):

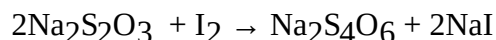
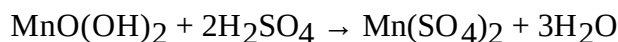
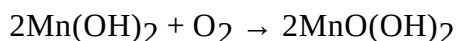
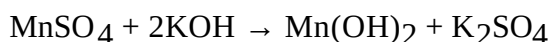
$$\text{Moles Na}_2\text{S}_2\text{O}_3 = V_{\text{Na}_2\text{S}_2\text{O}_3} \times \text{molarity}_{\text{Na}_2\text{S}_2\text{O}_3}$$

From the information given above, calculate the number of grams of O_2 in your 200 mL sample. From this, calculate the number of milligrams of O_2 per liter solution. Since a liter will weigh approximately 1000 g (the bulk of the solution is water), 1 mg/L is equivalent to 1 mg in 10^6 mg, or 1 million mg, of solution. Therefore, the number of milligrams of O_2 per liter is often referred to as parts per million (ppm).

Example 2

To a 200 mL water sample were added 0.5000 g KI, 2.000 mL of 2.000 M NH_2SO_4 and 2.000 mL of starch solution. The liberated iodine required 7.88 mL of 0.0251 M $\text{Na}_2\text{S}_2\text{O}_3$. Calculate the O_2 concentration in the sample in ppm.

Solution From analytic reactions



it is seen that each mole of O_2 requires 4 mol of $\text{Na}_2\text{S}_2\text{O}_3$.

$$\text{Moles Na}_2\text{S}_2\text{O}_3 = (\text{volume Na}_2\text{S}_2\text{O}_3) \times (\text{molarity Na}_2\text{S}_2\text{O}_3)$$

$$= (0.00788 \text{ L})(0.0251 \text{ mol/L})$$

$$= 1.98 \times 10^{-4} \text{ mol}$$

$$\text{Moles O}_2 = \frac{1}{4} \text{ moles Na}_2\text{S}_2\text{O}_3$$

$$= 4.95 \times 10^{-5} \text{ mol}$$

$$\text{Weight O}_2 = (4.95 \times 10^{-5} \text{ mol}) \times (32.0 \text{ g/mol})$$

$$= 1.58 \times 10^{-3} \text{ g}$$

$$\text{Concentration O}_2 = \frac{1.58 \text{ mg}}{0.200 \text{ L}} = 7.90 \text{ mg/L}$$

$$= 7.90 \text{ ppm}$$

A correction factor may be applied to your answer to correct for solution loss during the addition of manganous sulfate and sulfuric acid. This amounts to multiplication by 204/200 if these reagents were added to a 200 mL bottle.

Comparisons in Dissolved Oxygen Contents

The amount of oxygen dissolved in water depends not only upon the amount of chemical pollution but also upon such factors as water temperature and the atmospheric pressure above the water. At temperatures between 0°C and 39°C, the amount of O₂ that will be present in oxygen-saturated distilled water is given by the equation

$$\text{ppm dissolved O}_2 = \frac{(P - p) \times 0.678}{35 + T} = \text{SLDO}$$

where P is the barometric pressure in mm Hg, T is the temperature of the water in °C, and p is the vapor pressure of water at the temperature of the water. Calculate the saturation level (SL) for your water sample. The percent saturation is given by

$$\% \text{ SL} = \frac{100(\text{DO in ppm})}{(\text{SLDO in ppm})}$$

Calculate the percent SL for your sample.

Example 3

A water sample at 12°C and 652 mm Hg was found to contain 7.90 ppm O₂. Calculate the percent saturation of this sample.

Solution:

$$\text{SLDO} = \frac{(652 \text{ mm} - 10.5 \text{ mm})(0.678 \text{ ppm} - \text{ }^\circ\text{C} / \text{mm})}{(35 + 12)^\circ\text{C}}$$

$$= 9.25 \text{ ppm}$$

$$\% \text{ SL} = \frac{(100)(7.90 \text{ in ppm})}{9.25 \text{ ppm}} = 85.4\%$$